What is Momentum? Momentum Preamble

Synopsis: What is momentum? It is simply a quantity that is a product of mass and velocity: p = mv, where p is the momentum in kg m/s, m is the mass in kg, and v is the velocity (vector!) in m/s. So imagine a person of mass 50 kg, moving at 2 m/s. This person has momentum, p = (50 kg)(2 m/s) = 100 kg m/s. Note that 1) to have momentum, you must be moving or have a velocity, so anything at rest has no momentum (or zero momentum); 2) velocity is a vector quantity, and so is the momentum. That means the direction of movement is important. If moving to the east is positive, then the west is negative, and so, accordingly, momentum toward the east and momentum toward the west have the opposite sign (the east is, by convention, positive).

Momentum is a quantity that is as fundamental as energy, mass, velocity, position, temperature, time, and so on. You should think of it as a distinct entity as you think of these other quantities. It's just that you can quantify momentum if you know both the mass and the velocity of the object of interest.

And it's the quantity that Nature really cares about conserving. Just as energy and mass are conserved, i.e., never created or destroyed (so the total always remains the same), Nature behaves to ensure the momentum is conserved (we will learn about the conservation in greater detail soon).

Momentum is a useful quantity when dealing with dynamics (force and motion). For example, why would stopping a 200 kg football player running at 5 m/s be much harder than stopping a 20 kg toddler running at 5 m/s? The key quantity is neither just the mass nor just the velocity. It's the product of those two, the momentum. The greater the momentum, the more difficult it is to stop. In other words, you need a larger force to slow a larger momentum. You need to apply a large force to produce a significant change in momentum. This change is called impulse, which depends on the time duration of the force you apply and the amount of force being applied, i.e., the product of force and time (we will learn about the impulse-momentum relation in greater detail soon).

In this Note, I will introduce the definition of momentum and consider a few examples illustrating this fundamental concept.

Performance Expectations: Describe momentum using the momentum equation and calculate the momentum of objects.

For PHY 151 — with vectors: Treat momentum as the vector p = mv, write a bounce as a change of sign, and find the change in momentum as the vector difference $p_f - p_i$.

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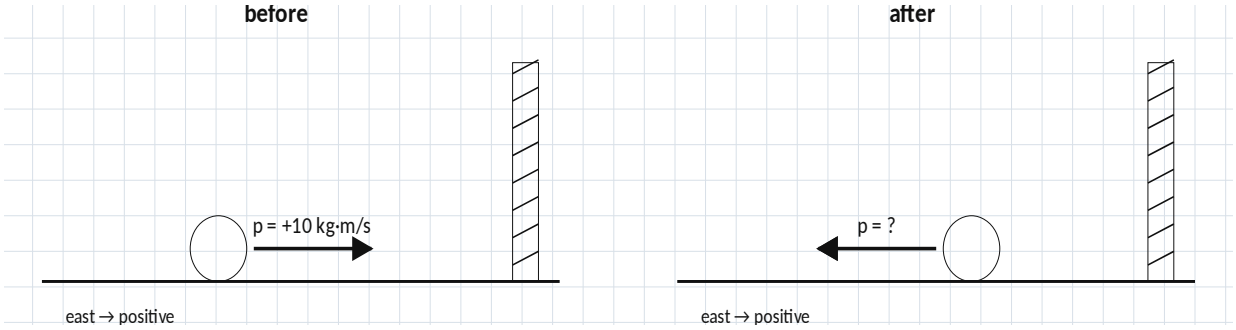
Instructions and Questions

1) Momentum is mass times velocity.
   i) Calculate the momentum of a 200 kg footballer running at 5 m/s and a 20 kg toddler running at 5 m/s.
   ii) If you double the velocity, what happens to the momentum?
   iii) If you triple the mass, what happens to the momentum?

2) Momentum has a direction.



   i) A bouncy ball moving toward the east has a momentum of 10 kg m/s. The ball bounces off the wall and now moves in the opposite direction. What is its new momentum?
   ii) The answer to this question requires a little bit of careful thinking. How much has the ball's momentum changed after bouncing off the wall? (Hint: It's not zero. Take the difference between momentum after and momentum before, and be careful with the signs of momenta.)

Before the wall	After the wall

3) A slow train and a fast bullet.
   i) A freight train has a mass of 5000 tons (1 ton = 1000 kg). It is lumbering along at a snail's pace, at 0.03 mph. Calculate the momentum in kg m/s.

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ii) A bullet with a mass of 25 g is fired from a rifle. It travels at 3000 km/h. Calculate the momentum in kg m/s.

iii) Based on your calculations above, discuss why bullet-proof jackets can stop a bullet, but one has zero chance of stopping a train. (Imagine the same momentum in a 200 kg football player – how fast would the player be going?)

Momentum as a vector.

p = mv is a vector equation: the momentum points the way the velocity points, and a component form $p_x = mv_x$, $p_y = mv_y$ is available whenever the motion is not along one line. A bounce off a wall is a change of sign in one component, and the change in momentum is the vector difference $\Delta p = p_f - p_i$, which is why a ball that comes back with the same speed has changed its momentum by twice its size.

i) A 0.50 kg ball hits a wall at 6.0 m/s, moving at 30° to the wall's normal, and leaves at the same speed and angle. Write p_i and p_f in components (x along the normal) and find Δp.

ii) Say in one sentence why only the normal component changed.

4) Looking ahead.

Momentum is the second conserved quantity of the module. Next we watch it being conserved — carts that bounce and carts that stick — and write the equation that says the total before equals the total after.

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